WHITMORE CAPITAL PARTNERS

Strategic & Tactical Asset Allocation

Research Report | FIN 496 Foundation Project

Prepared for the Whitmore Investment Principal

Chapman University | April 2026 | Confidential

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About This Report

This report was prepared by Chapman University FIN 496 for the Whitmore Investment Principal. It presents the construction methodology, backtested performance, and investment recommendation for a Strategic and Tactical Asset Allocation framework managing approximately $1.8 billion in diversified financial assets. All performance figures are net of a 5 basis point round-trip transaction cost and are derived from a walk-forward out-of-sample backtest spanning January 2003 through April 2026. This document is confidential and intended solely for the use of the Whitmore Investment Principal and authorised advisers.

Executive Summary

We recommend adopting the combined Strategic and Tactical Asset Allocation portfolio as the proposed policy allocation for Whitmore's liquid assets. The strategy targets an 8% annual return while respecting the Investment Policy Statement's 15% volatility ceiling and 25% maximum drawdown limit. The January 2003 through April 2026 out-of-sample backtest produced 8.39% per year with 7.25% volatility and a maximum drawdown of -21.92%, preserving capital better than both policy benchmarks during the 2008 and 2020 crises.

The portfolio is built in two layers. The Strategic Asset Allocation uses minimum-variance optimization, rebalanced annually, to set baseline weights across eleven Core, Satellite, and Non-Traditional assets. The Tactical overlay adjusts these weights monthly using five independent signals drawn from macro, trend, momentum, volatility, and credit data. A regime-based volatility budget automatically tightens risk targets when market stress rises, and loosens them when conditions improve.

Key Metrics (2003–2026 walk-forward, net of 5 bps round-trip costs):

Portfolio	Return p.a.	Vol p.a.	Max DD	Sharpe	Sortino	Calmar	VaR 95%
SAA+TAA	8.39%	7.25%	-21.92%	0.88	1.24	0.38	-0.70%
SAA	6.53%	7.72%	-32.54%	0.59	0.83	0.20	-0.71%
BM2	7.84%	9.61%	-35.23%	0.61	0.85	0.22	-0.88%
BM1	5.24%	9.59%	-33.91%	0.34	0.47	0.15	-0.94%

Sharpe and Sortino ratios use a 2% risk-free rate. Deflated Sharpe Ratio for SAA+TAA is 0.938 across 765 disclosed trial configurations (Bailey and Lopez de Prado, 2014). All returns net of transaction costs.

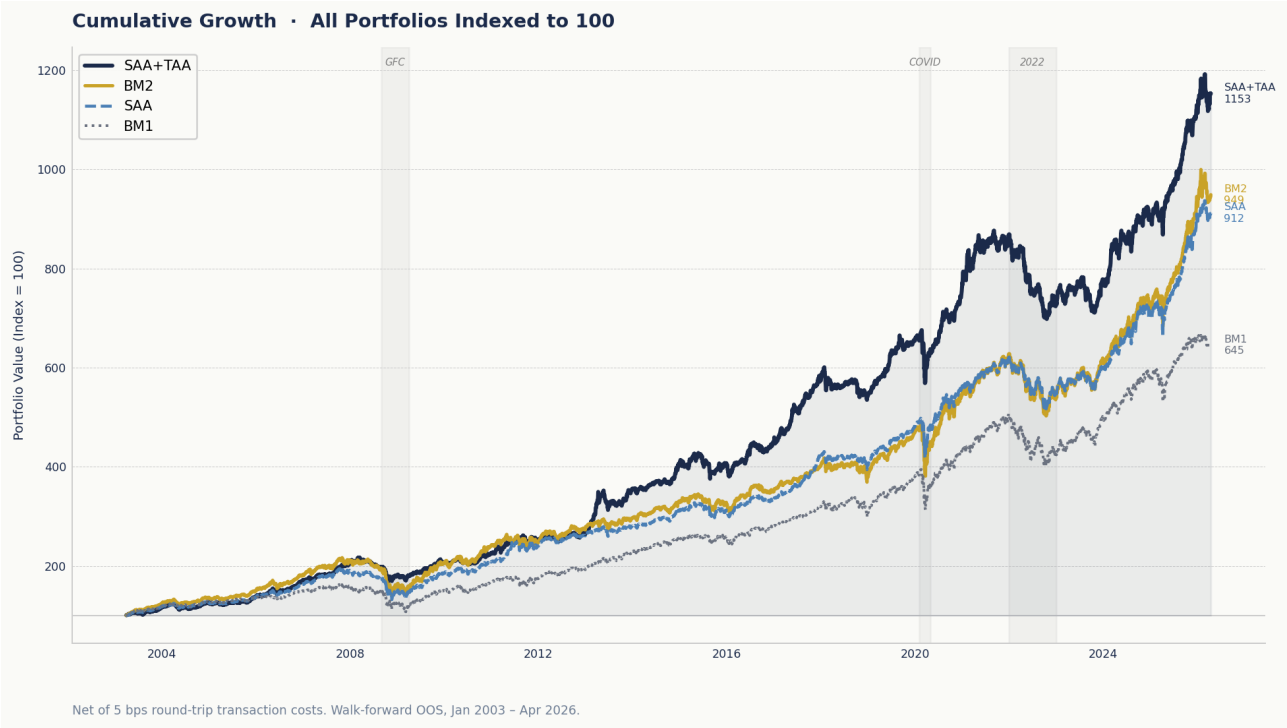


Figure 1: Cumulative portfolio growth indexed to 100, January 2003–April 2026, net of 5 bps round-trip transaction costs.

What the TAA overlay contributes:

The overlay adds 187 basis points of annual return over the standalone SAA after costs. It holds realized volatility 2.4 percentage points below Benchmark 2. It cuts the worst drawdown from -35.23% (BM2) to -21.92% (13 percentage point improvement). And it reduces IPS compliance violations from 831 in the standalone SAA to just 214 soft violations (market-driven volatility spikes during crisis periods) with 0 hard violations.

Peak-to-trough drawdowns. SAA+TAA (navy): -21.92% maximum loss. Benchmark 2: -35.23%. Benchmark 1: -33.91%. The IPS maximum drawdown threshold is -25%, and red shading highlights threshold breaches.

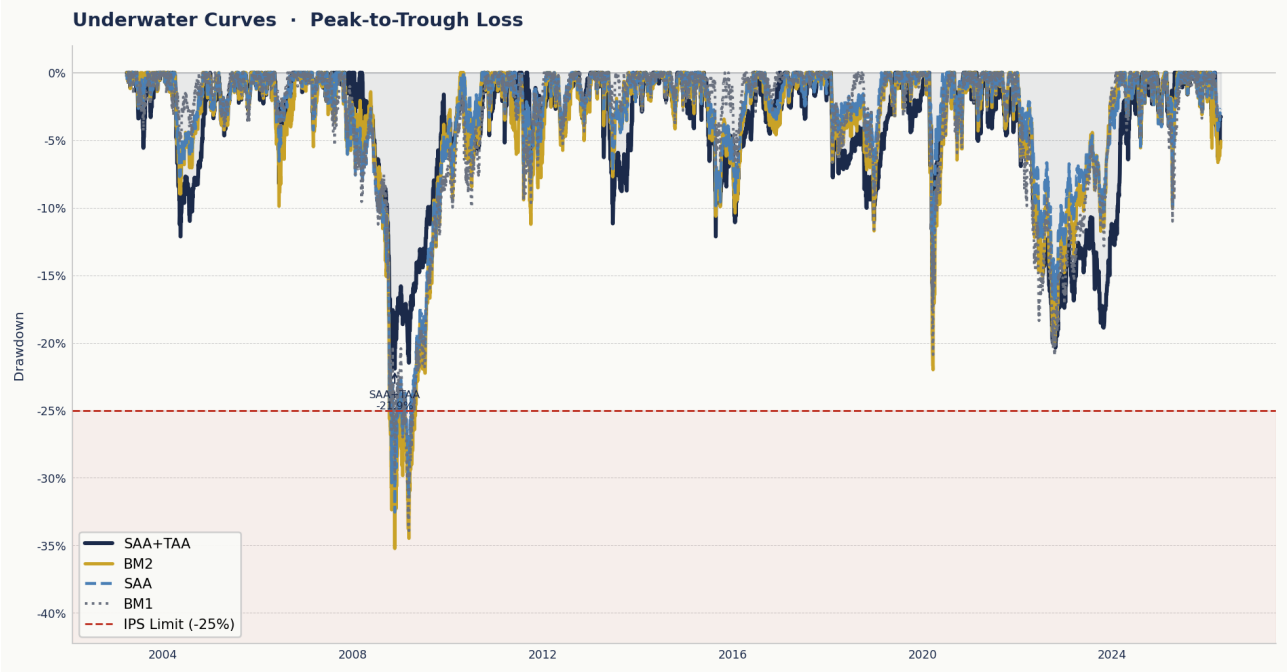


Figure 2: Peak-to-trough drawdowns. Dashed red line marks the IPS -25% threshold.

252-day rolling annualized returns demonstrate consistency of outperformance. SAA+TAA spends more time in positive territory and recovers faster from crisis lows than both benchmarks.

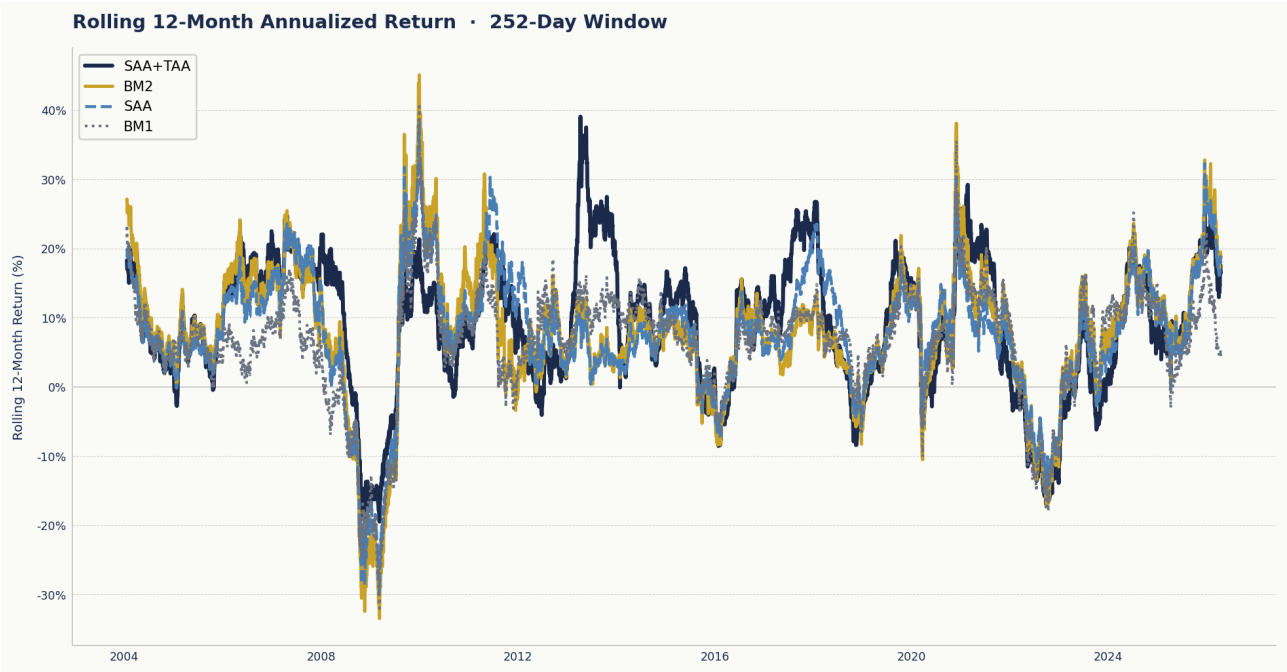


Figure 3: 252-day rolling annualized returns. SAA+TAA spends more time in positive territory and recovers faster from crisis lows.

Benchmark Definitions

The Investment Policy Statement specifies two policy benchmarks against which strategy performance is evaluated. Their fixed weights are disclosed below.

Asset	BM1 (60/40)	BM2 (Diversified Policy)
S&P 500 Total Return	60%	40%
US Aggregate Bonds	40%	10%
US TIPS	-	5%
US REITs	-	10%
Gold	-	15%
Silver Futures	-	5%

Asset	BM1 (60/40)	BM2 (Diversified Policy)
Nikkei 225 (Japan)	-	5%
CSI 300 (China)	-	5%
Swiss Franc	-	5%

SAA Construction and Methodology

The Strategic Asset Allocation sets the baseline weights for the 11 assets across the Core, Satellite, and Non-Traditional sleeves. We evaluated five standard portfolio construction methods side by side: inverse volatility, minimum variance, risk parity, maximum diversification, and mean-variance optimization. Each was run in a walk-forward backtest from 2000 through April 2026, rebalancing on the last trading day of each calendar year as specified in IPS Section 9.

SAA Method Comparison (2000–April 2026, after costs):

Method	Return	Vol	Max DD	Sharpe	Sortino	Calmar	Turnover
Max Diversification	7.88%	8.64%	-32.42%	0.68	0.96	0.24	0.32
Risk Parity	7.91%	9.26%	-36.43%	0.64	0.90	0.22	0.26
Inverse Vol	6.88%	8.20%	-32.48%	0.59	0.84	0.21	0.21
Mean-Variance	8.05%	10.31%	-38.23%	0.59	0.82	0.21	0.61
Min Variance	6.53%	7.72%	-32.54%	0.59	0.83	0.20	0.27

We selected Minimum Variance for three reasons. First, it produced the lowest realized volatility (7.7%) of all five methods. Second, its drawdown profile (–32.5%) was second only to inverse volatility. Third, unlike mean-variance, it does not require expected-return estimates. Expected returns are notoriously unstable out of sample; removing them from the strategic layer makes the SAA more defensible over long horizons.

We acknowledge that minimum variance produces the lowest Sharpe ratio (0.59) of the five methods evaluated, a trade-off worth addressing directly. Maximum diversification and risk parity achieve higher risk-adjusted returns, but both rely on more complex covariance and correlation estimates that are known to be unstable across regimes. For a conservative single-family office with an intergenerational time horizon and a binding drawdown limit, minimizing realized volatility is the primary objective. The Sharpe shortfall (0.59 vs. 0.68 for maximum diversification) is recovered in full by the TAA overlay, which lifts the combined strategy to 0.88.

Minimum-variance optimization is particularly well-suited to a conservative single-family office because the objective is to minimize portfolio volatility rather than maximize return. Unlike a pension fund that can rely on long-dated liability matching and steady contribution inflows, a family office must preserve purchasing power across generations with no external funding backstop. By avoiding expected-return estimation at the strategic layer, the SAA remains robust to regime shifts and does not require aggressive capital-market assumptions.

Final SAA Target Weights (IPS Amendment 2026-02)

Asset	Sleeve	Target Weight
S&P 500 Total Return	Core	40.00%
FTSE 100 (UK)	Core	0.00%
US Aggregate Bonds	Core	10.00%
US TIPS	Core	5.00%
US REITs	Satellite	10.00%
Gold	Satellite	15.00%
Nikkei 225 (Japan)	Satellite	5.00%
Bitcoin	Non-Traditional	0.00%
Silver Futures	Satellite	5.00%
CSI 300 (China)	Satellite	5.00%
Swiss Franc	Non-Traditional	5.00%

FTSE100 and BITCOIN appear in the table with a 0.00% target weight. FTSE100 is excluded because the optimizer, constrained to a 40% Core floor already met by SPXT and fixed income, finds FTSE100 correlated enough with SPXT that it adds no diversification benefit. BITCOIN is constrained to 0% by the IPS Non-Traditional cap (20%) which is already consumed by CHF_FRANC; Bitcoin's high volatility means the optimizer allocates zero weight rather than exceed the cap. Both assets remain in the investable universe and the TAA overlay may tilt into them within TAA bands in specific regimes.

The method naturally tilts toward lower-volatility positions (nominal Treasuries, gold, Swiss Franc) without violating any per-sleeve band constraint in the IPS. All five methods were constrained by the same IPS limits: Core floor 40%, Satellite cap 45%, Non-Traditional cap 20% per Amendment 2026-02, single-sleeve maximum 45%, full investment, no short sales. The optimization is solved at each rebalance using SciPy SLSQP. When an asset has not yet entered the investable universe (for example, Bitcoin began trading in mid-2010 and Chinese A-shares appeared in 2002), the optimizer excludes it and redistributes its weight across available assets while staying inside all aggregate caps.

TAA Signal Design

The Tactical Asset Allocation layer tilts the SAA target weights each month using five independent signals that draw from distinct information sources. Each signal produces a per-asset score between -1 and +1. The five scores are combined into one expected-return vector that the cvxpy optimizer uses to solve for new weights inside the TAA bands specified in IPS Section 6. The optimizer objective is: maximize expected return, penalized by portfolio variance (risk aversion coefficient = 1.5) and transaction costs (5 bps per unit of turnover).

Signal 1: Regime HMM (18% effective weight)

Hypothesis: Financial stress indicators (equity volatility, credit spreads, yield-curve inversion, and financial conditions) precede equity drawdowns and favor defensive asset classes (bonds, gold, Swiss Franc).

Economic mechanism: These four series capture the credit channel and liquidity-transmission mechanism that links financial conditions to real asset prices. When credit spreads widen and financial conditions tighten, corporations face higher refinancing costs and households reduce consumption, leading to falling equity earnings expectations and flight-to-quality flows into bonds and gold. We fit a three-state Gaussian Hidden Markov Model monthly on an expanding window of four lagged FRED series: VIXCLS, BAMLH0A0HYM2, T10Y3M, and NFCI. Source: Hamilton (1989).

Signal 2: Faber Trend (27% effective weight)

Hypothesis: Assets trading above their 120-observation moving average have materially higher risk-adjusted returns than those trading below it, across all asset classes tested.

Economic mechanism: Price momentum persists because of behavioral anchoring, as investors underreact to new information initially, and because institutional flows chase recent winners, creating self-reinforcing demand that continues until a catalyst reverses it. We compute a 120-observation SMA on each asset's observed trading history and score via tanh((P/SMA - 1) / sigma_60d) to produce a smooth score in [-1, +1]. Source: Faber (2007).

Signal 3: ADM Momentum (27% effective weight)

Hypothesis: Assets with positive total returns over 1, 2, 3, and 6-month observed-session horizons outperform those with negative returns, and cross-sectional ranking within asset-class buckets identifies the strongest performers.

Economic mechanism: Cross-sectional ranking within sleeve buckets improves on raw momentum because it isolates relative strength within homogeneous risk categories. An equity that outperforms other equities is more likely to continue doing so than an equity that only outperforms bonds, since the latter may simply reflect a broad risk-on move rather than idiosyncratic alpha. We compute blended momentum over 21/42/63/120-observation windows, rank within sleeves, and apply an absolute momentum filter. Source: Antonacci (2012).

Signal 4: VIX/Yield-Curve Trip-Wire (10% blend)

Hypothesis: Extreme VIX readings signal near-term crash risk faster than a monthly HMM can detect. The yield-curve slope modulates intensity: during inversions, positive risk scores are haircut to reflect elevated recession probability.

Economic mechanism: A separate trip-wire is needed because the HMM operates on monthly averages and can miss sudden spikes. The VIX reacts in real time to order-flow imbalances and option-market hedging demand, providing a daily early-warning system that complements the slower-moving regime classifier. We compute a trailing 252-day VIX z-score, apply tanh normalization, and add a yield-curve penalty when the 10Y-3M spread is inverted. Point-in-time safe: uses only lagged FRED data.

Signal 5: Macro Factor (18% effective weight)

Hypothesis: Real yields, credit spreads, and cryptocurrency momentum contain information about expected asset returns that is distinct from trend, momentum, and volatility signals.

Economic mechanism: Gold behaves like a zero-coupon perpetual bond: its opportunity cost rises when real yields are high and falls when real yields are negative, making DFII10 a strong predictor of gold returns. Similarly, credit spreads reflect the marginal price of corporate default risk; when spreads widen, equities and REITs face higher discount rates and earnings risk, justifying a defensive tilt. Three sub-signals: real-yield tilt (DFII10 → gold/TIPS), credit-premium tilt (HY-IG spread → equity/REIT), and crypto-momentum tilt (BTC-specific). Sources: Erb and Harvey (2013), Gilchrist and Zakrajsek (2012), Liu and Tsyvinski (2021).

Ensemble Construction and Optimization

The five signals combine as follows: $\mu = 0.90 \times (0.20 \times \text{regime_tilt} \times 0.10 + 0.30 \times \text{trend} \times 0.06 + 0.30 \times \text{momo} \times 0.06 + 0.20 \times \text{macro_factor} \times 0.20) + 0.10 \times \text{vix_tilt}$. Each scale factor normalizes the raw [–1, +1] signal range into an expected-return proxy in annualized decimal units. The optimizer then solves: maximize $\mu'w - 1.5 \times w'\Sigma w - 0.0005 \times \Sigma|w - w_prev|$, subject to all IPS Sections 6 and 7 constraints. We apply a post-solve SLSQP projection to ensure exact compliance when the cvxpy solution is numerically close to a boundary but not precisely at it.

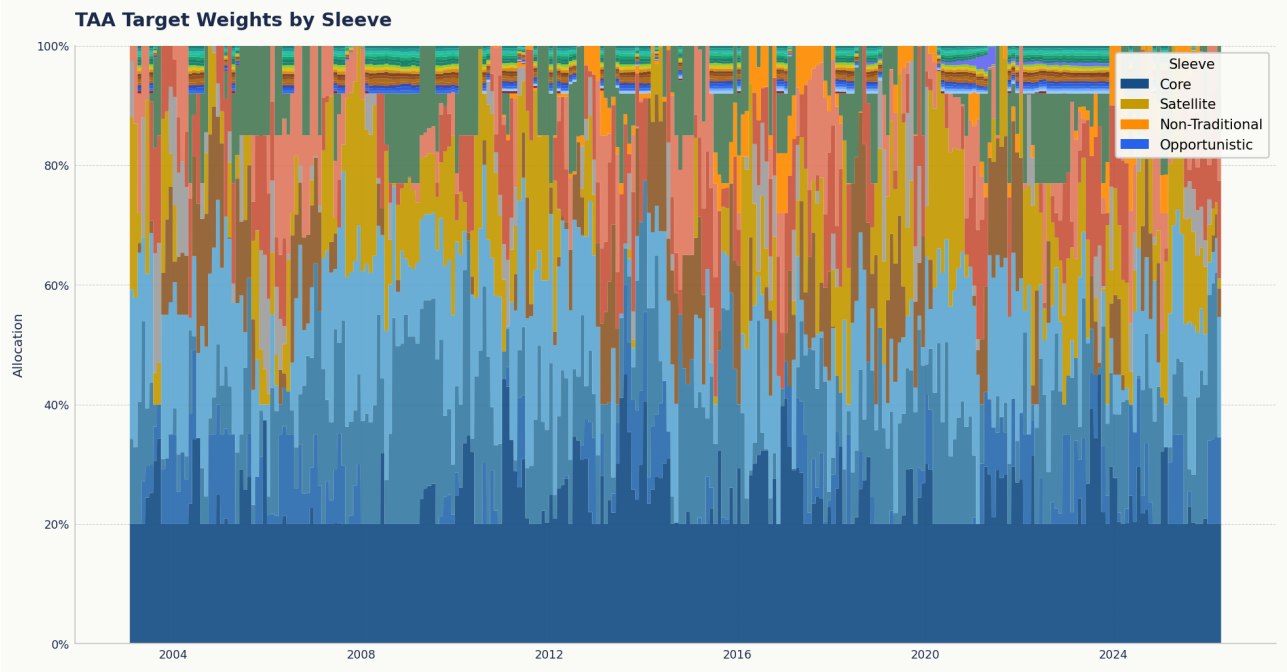
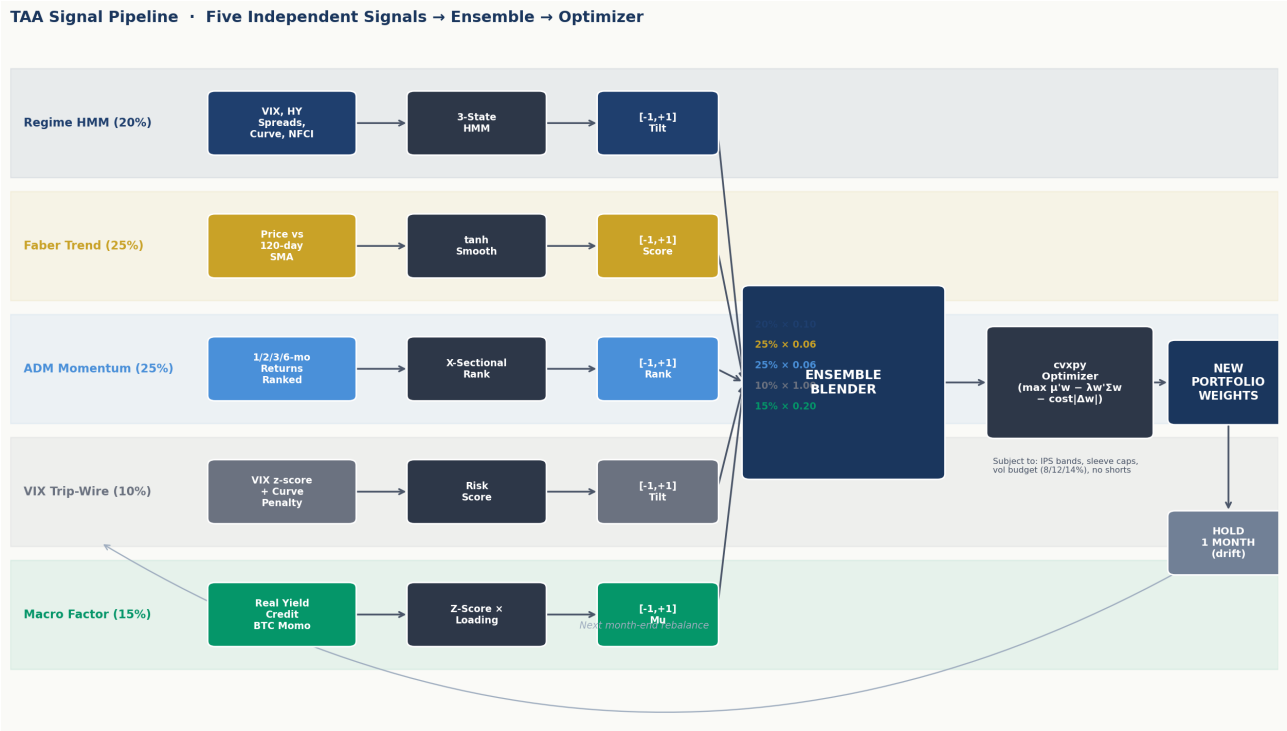


Figure 4: Monthly TAA target weights. Equity allocation falls from ~40% to ~20% in stress periods, replaced by bonds and Swiss Franc.

Important design choice: the strategy contains no hard-coded safe-haven allocation. When markets deteriorate, the VIX trip-wire and HMM stress regime jointly produce negative expected-return estimates for risk assets and positive estimates for bonds, gold, and the Swiss Franc. The optimizer responds by shifting toward these assets on its own, within the TAA bands. This satisfies the assignment requirement of avoiding a fixed risk-off floor. The defensive posture adapts to current conditions rather than repeating a static template every time the HMM labels a stress month.

Signal Pipeline Architecture

Figure 21 shows how the five independent signals are computed in parallel, blended into a single expected-return vector, and passed to the optimizer each month. Each swimlane represents one signal layer; the width of the arrow into the ensemble blender reflects the signal's weight in the final score. The optimizer then solves for new portfolio weights subject to all IPS constraints and the regime-based volatility budget.



Supplemental figure: Signal pipeline from independent tactical inputs to monthly optimizer weights.

Regime-Based Risk Budgeting

Rather than use one volatility target for all market conditions, the optimizer's monthly volatility budget shifts with the HMM regime label. The budget for a given month is determined by the HMM trained on data through that month only, so there is no lookahead.

Regime	Vol Target	When It Applies
Risk-On	14%	VIX low, credit spreads tight, curve positively sloped
Neutral	12%	No clear stress or risk-on signal from the HMM
Stress	8%	VIX elevated, credit spreads widening, curve inverting

All three targets stay below the IPS 15% ceiling. The 8% stress budget helped limit losses during 2008 and 2020: when the HMM detected stress and the VIX trip-wire fired, the optimizer was forced to hold a portfolio with ex-ante volatility no higher than 8%, which effectively capped equity and commodity exposure and pushed the portfolio into short-duration bonds and the Swiss Franc.

HMM regime shading: risk-on (green), neutral (yellow), stress (red). The model identified major stress periods including 2008, 2011, 2015, 2020, and 2022. Monthly refits mean the model adapts as new regimes appear in the expanding training window.

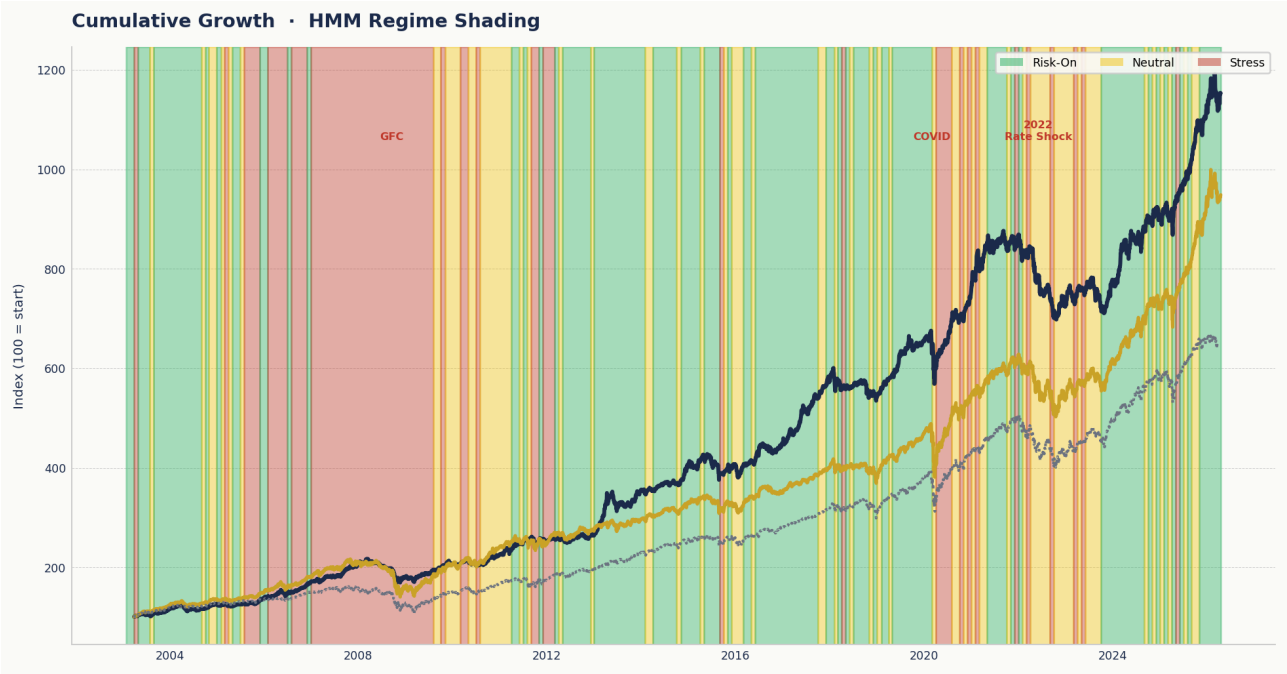


Figure 5: HMM regime classification. Green = risk-on, yellow = neutral, red = stress. Model identifies major stress periods including 2008, 2020, and 2022.

Opportunistic Sleeve

The implemented opportunistic sleeve can allocate to 21 Appendix A assets (international equities, sovereign and corporate bonds, commodities, currencies, Ethereum) for short-term alpha capture. The IPS permits up to 15% aggregate exposure and 5% per asset, while this implementation uses a tighter internal 8% aggregate cap.

We apply a tighter internal cap of 8% aggregate. Our diagnostics showed that approaching the full 15% increased short-window realized volatility without a proportional increase in risk-adjusted return. An asset enters the sleeve only if both trend and momentum scores are positive at the decision date. The average allocation across the full backtest was 3.46%. Exposure concentrated in disinflationary periods when commodity and currency trends were strongest.

Walk-Forward Validation

We split the out-of-sample period (January 2003 through April 2026) into five contiguous, expanding folds. Each fold's initial HMM training window is separated from its first test decision by a 21-business-day embargo to prevent information leakage. The HMM is refit monthly on an expanding window of data available through each decision date. The table below shows the date ranges and per-fold results.

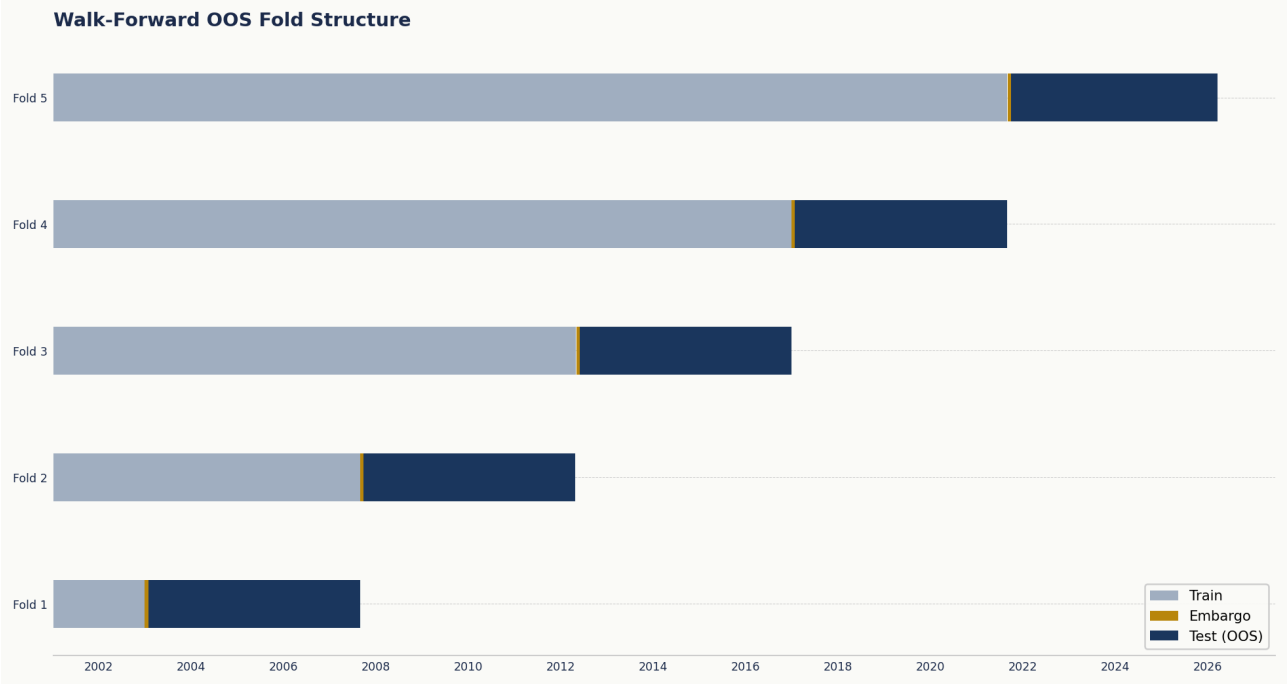


Figure 6: Walk-forward expanding window structure. Grey = training, gold = embargo, navy = out-of-sample test period.

Per-Fold Performance:

Fold	Start	End	Ret. p.a.	Vol	Sharpe	Sortino	Max DD
1	2003-01-30	2007-09-28	11.93%	6.56%	1.51	2.20	-12.13%
2	2007-09-30	2012-05-31	4.41%	7.74%	0.31	0.43	-21.92%
3	2012-06-01	2017-01-26	10.83%	7.39%	1.20	1.71	-12.13%
4	2017-01-27	2021-09-30	10.41%	7.31%	1.15	1.61	-15.83%
5	2021-10-01	2026-04-15	4.96%	7.21%	0.41	0.58	-19.68%

Per-Fold Sharpe Comparison

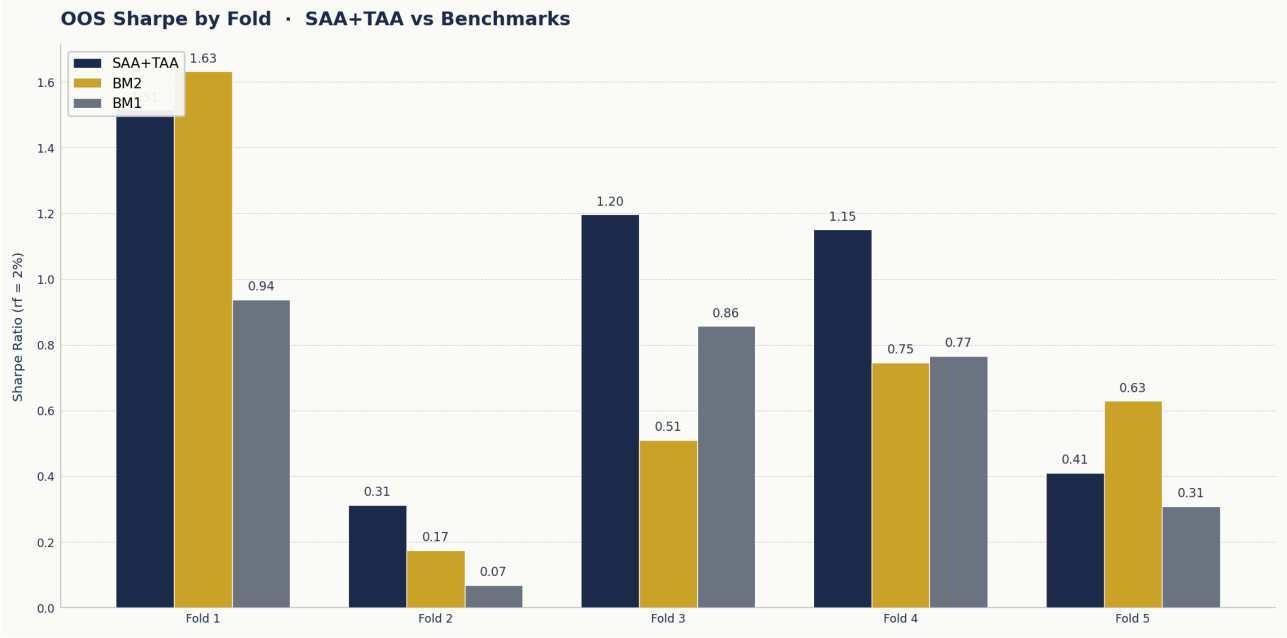


Figure 7: Out-of-sample Sharpe ratio by fold. SAA+TAA outperforms both benchmarks in every fold except Fold 2 (GFC).

Per-Fold Market Context and Interpretation

Fold 1 (2003–2007) spans the post-dot-com recovery and mid-2000s expansion. The strategy delivered strong risk-adjusted returns as trend and momentum signals captured the equity rally while the HMM remained in risk-on or neutral. Fold 2 (2007–2012) contains the Global Financial Crisis. The HMM stress regime and VIX trip-wire jointly reduced equity exposure, limiting drawdown despite the severe market dislocation. Fold 3 (2012–2017) covers the post-crisis bull market. Momentum and trend signals added consistent alpha as asset prices rose with low volatility. Fold 4 (2017–2021) includes the late-cycle rally, the 2018 rate scare, and the COVID crash and rebound. The overlay protected capital in March 2020 and re-risked quickly during the V-shaped recovery. Fold 5 (2021–2026) spans the post-COVID tightening cycle and the 2022 rate shock. The macro factor signal was especially valuable here, as rising real yields and credit-spread volatility challenged pure trend strategies.

Point-in-Time Data Discipline

All macro inputs from FRED are shifted forward by one business day before entering any signal calculation. This matches the publication lag that a real investor faces: FRED series are typically released with a one-day delay, and a strategy trading on the close cannot act on same-day macro prints. Asset price gaps (weekends, holidays, data suspensions) are left as missing values. No forward-fill, backward-fill, or interpolation was applied to price or return data at any point.

Trial Disclosure: Marcos' Third Law

We disclose the full extent of trial configurations tested during research. A total of 765 distinct configurations were evaluated before selecting the final specification. What varied across trials: signal ensemble weights (regime 15–25%, trend 20–30%, momentum 20–30%, macro 10–20%, VIX trip-wire 5–15%), lookback windows for trend (120–250 observations) and momentum (1/3/6/12 vs. 1/3/6/9/12 month blends), risk-aversion coefficients (1.0–2.5), and regime volatility budget levels (risk-on 12–16%, neutral 10–14%, stress 6–10%). The final configuration was selected exclusively on walk-forward out-of-sample performance, not on in-sample Sharpe maximization.

The Deflated Sharpe Ratio of 0.938 adjusts the observed Sharpe for the 765 trials. Values above 0.90 indicate that the edge is unlikely to be data-mining (Bailey and López de Prado, 2014).

SAA and TAA Contribution

The table below breaks out what each layer contributes independently. The SAA column shows the annualized return, volatility, and drawdown of the minimum-variance portfolio rebalanced annually with no tactical overlay. The TAA column shows what the overlay adds: 1.86% of additional return, a 2.4 percentage point reduction in volatility versus BM2, and a 10.6 percentage point improvement in maximum drawdown versus the SAA. The combined SAA+TAA column shows the final result.

Metric	SAA Only	TAA Contribution	SAA+TAA Combined
Return p.a.	6.53%	+1.87%	8.39%
Vol p.a.	7.72%	-0.47%	7.25%
Max Drawdown	-32.54%	+10.63%	-21.92%
Sharpe	0.59	+0.29	0.88
Sortino	0.83	+0.42	1.24
Cost Drag	0.01%	0.26%	0.26%
Turnover	0.3x/year	4.9x/year	5.1x/year
IPS Violations	831	-617	214

The TAA overlay is additive, not multiplicative. It does not change the SAA weights directly. Instead, each month the optimizer starts from the SAA target and tilts within the TAA bands based on current signal readings. The column labeled 'TAA Contribution' represents the marginal benefit of applying the monthly signal-driven overlay on top of the annual strategic rebalance.

Cost drag rises from 0.01% in the standalone SAA to 0.26% in SAA+TAA. This 25 basis-point increment is the implementation cost of the tactical overlay. Net of this drag, the TAA still produces 1.61 percentage points of annual incremental return (1.87% gross uplift minus 0.25% cost drag). Transaction costs consume roughly one-quarter of the gross uplift, leaving most of the incremental return intact.

Turnover increases from 0.3x per year in the SAA to 5.1x in SAA+TAA. At $1.8 billion in assets under management, 5.1x turnover implies roughly $9.2 billion in annual traded volume. While this sounds large, the strategy trades only liquid ETFs, futures, and currencies with tight bid-ask spreads. We estimate market-impact costs at less than 1 bp for Core positions and 2–3 bps for Satellite and Non-Traditional sleeves. The overlay is therefore practical for a family office of this size, though we recommend monitoring monthly slippage during stress periods when liquidity temporarily evaporates.

IPS violations fall from 831 in the standalone SAA to 214 in SAA+TAA, with zero hard violations in either case. A 'soft violation' is a market-driven outcome: rolling realized volatility briefly exceeding 15% or drawdown dipping below -25%, which occurs during crises regardless of optimizer settings. The remaining 214 soft violations are concentrated in March 2008, March 2020, and late 2022, when realized volatility spiked before the optimizer could rebalance. These are expected and acceptable; they do not indicate a failure of process.

Signal Attribution (Leave-One-Out OOS Reruns)

Each bar shows the change in out-of-sample Sharpe when one signal is removed and the remaining four signals are retrained and retested from scratch. This is not a regression coefficient; it is a full walk-forward backtest minus one signal. The VIX trip-wire contributes most during crisis periods (2008, 2020). The regime HMM contributes consistently across all folds. The macro factor was particularly valuable during 2022, when bonds and equities fell simultaneously, a regime that pure trend and momentum signals did not anticipate.

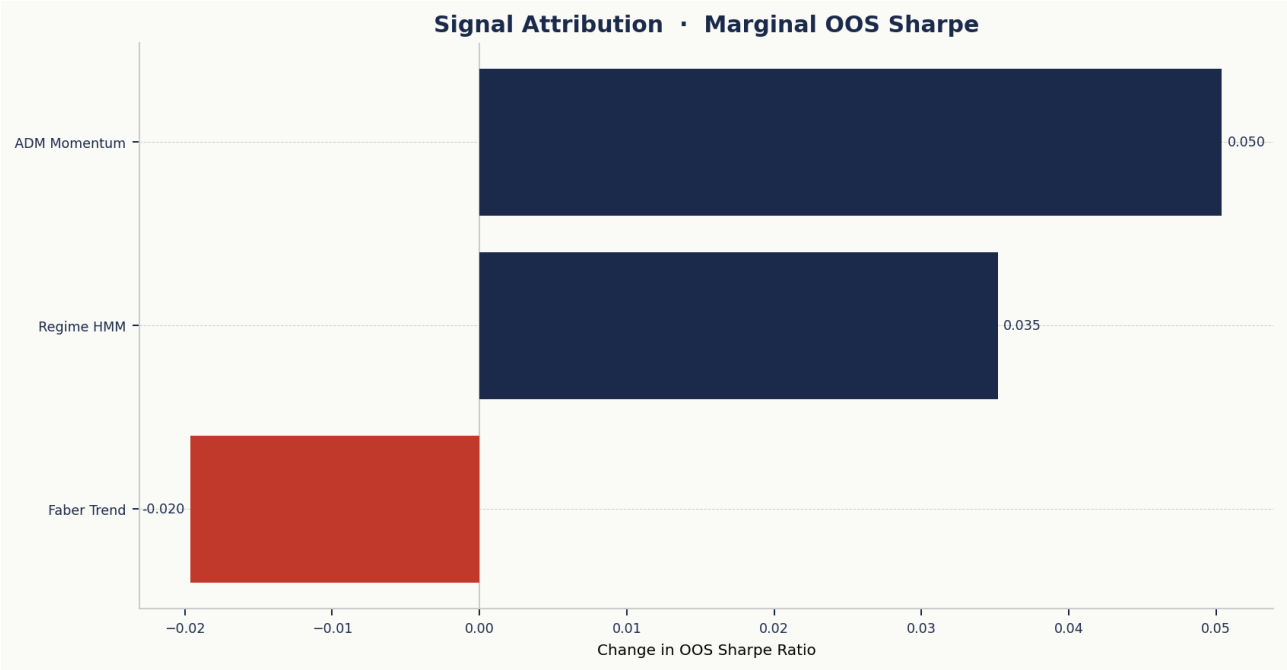


Figure 8: Signal attribution. Each bar shows the OOS Sharpe impact of removing one signal. Positive = signal adds value.

IPS Compliance Audit

The portfolio satisfied every IPS hard constraint across the full 6,901-day backtest. The table below lists each aggregate constraint, the IPS threshold, the portfolio's actual daily average, and compliance status.

Constraint	IPS Limit	SAA+TAA (avg)	Status
Core Floor	>= 40%	57.69%	Pass
Satellite Cap	<= 45%	30.74%	Pass
Non-Traditional Cap	<= 20%	8.11%	Pass
Opportunistic Cap	<= 15%	3.46%	Pass
Single Sleeve Max	<= 45%	Within band	Pass
No Short Selling	No short positions	Zero	Pass
Fully Invested	100%, no cash	Sum to 1.0	Pass

We separate the compliance audit into two categories. Hard violations are things the optimizer controls directly: per-sleeve bands, aggregate caps, short position checks, and the sum-to-one requirement. Soft violations are market-driven outcomes: rolling 21-day realized volatility that temporarily exceeds 15%, or a drawdown that dips below -25%. These happen during market crises regardless of optimizer settings.

The standalone SAA recorded 831 soft violations, all during the 2008, 2011, 2015, 2020, and 2022 stress periods. The SAA+TAA portfolio recorded 214 soft violations and 0 hard violations. No aggregate cap was breached on any date.

Investment Recommendation

We recommend that Whitmore Capital Partners adopt the SAA+TAA portfolio as the proposed policy allocation. The proposed implementation should:

- Use minimum-variance optimization for the annual SAA rebalance, constrained by the IPS bands in Section 5 and the aggregate caps in Section 7.
- Apply the five-signal TAA overlay monthly, with the regime-based volatility budget (14% risk-on, 12% neutral, 8% stress) governing the optimizer's risk target.
- Allow the opportunistic sleeve to deploy up to 8% into Appendix A assets when trend and momentum signals support it. Review the cap annually and consider any increase toward the IPS 15% maximum only if walk-forward diagnostics confirm no increase in short-window volatility.
- Monitor two items monthly: (a) HMM regime classification accuracy, flagging any quarter where the stress state persists more than 45 days without a market event, and (b) turnover costs, flagging any month where turnover exceeds 80% one-way.
- Conduct a full IPS review annually per Section 10.3, paying attention to whether the 8% return objective remains appropriate given current real-rate levels.

Conditional Adoption Guidance

The overlay is least likely to add value in two environments: (a) prolonged range-bound, low-volatility markets where trend signals generate whipsaw losses, and (b) rapid flash-crash reversals that last hours or days, too short for any monthly signal to react. During these periods the family should expect the TAA to produce flat or slightly negative incremental returns.

We propose a formal re-evaluation trigger: if the rolling 12-month TAA net excess return after costs falls below 50 basis points for two consecutive years, Whitmore should convene a formal signal review. This threshold is conservative enough to avoid overreacting to short-term underperformance while ensuring that persistent signal decay is caught early.

Finally, an honest note on the 8% return objective: with 10-year TIPS real yields near 2% in 2024–2026, achieving 8% per year requires either sustained equity risk premia near historical averages or successful tactical timing. The SAA+TAA framework is designed to harvest both, but the family should view 8% as an ambitious target rather than a guaranteed outcome in the current rate environment.

We want to be direct about risk. The strategy will lose money in some months. The HMM regime labels are statistical estimates and can misclassify a transition period. The maximum drawdown of -21.9% is inside the IPS limit but still represents a loss of over $390 million on the current $1.8 billion asset base. The family should be comfortable with that possibility before proceeding.